5. Inclusion–exclusion and applications

Combinatorics
MATH 31003-01 - Fall 2026

Suppose we want to count the students that are taking classes in math: algebra or geometry. Adding the two course enrollments seems natural, but that sum counts every student in both courses twice. With three or more courses, an individual student may appear in several different intersections, so the correction requires more care.

The sum rule counts the union of mutually disjoint sets by adding their sizes. When sets overlap, however, simply adding their cardinalities overcounts elements that belong to multiple sets. For two sets, subtracting the size of their intersection corrects this double count. For three or more sets, elements may belong to one, two, or more sets. The systematic pattern to account for this is known as the **principle of inclusion–exclusion**.

After studying these notes, you will be able to do the following.

- Explain and apply the two-set and three-set inclusion–exclusion formulas.
- State the general principle of inclusion–exclusion for m finite sets and compute sizes of unions.
- Use indicator functions to prove the inclusion–exclusion formula.
- Express problems with multiple constraints using the complement (violation-set) form.
- Count integer solutions to linear equations with upper bounds by combining inclusion–exclusion with stars and bars.
- Count surjective functions and strings that use every symbol of an alphabet.
- Count permutations avoiding prescribed substrings using block reduction.
- Derive the formula for derangements D_n and count permutations with a specified number of fixed points.

5.1 The geometry of overlapping sets

Let A and B be finite subsets of a universal set U. The union $A \cup B$ consists of all elements belonging to A, to B, or to both. When $A \cap B = \emptyset$, the sum rule gives $|A \cup B| = |A| + |B|$. When $A \cap B \neq \emptyset$, every element $x \in A \cap B$ contributes 1 to $|A|$ and 1 to $|B|$, so the sum $|A| + |B|$ counts each element of $A \cap B$. Subtracting $|A \cap B|$ removes this extra copy.

Theorem 5.1.1 (Inclusion–exclusion for two sets)**.** For any finite sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Example 5.1.2 (Students enrolled in two courses)**.** In a department of mathematics, 30 students take abstract algebra, 25 take geometry, and 8 take both courses. How many students take at least one of these two courses?

Solution: Let A be the set of students taking algebra and G the set of students taking geometry. The set of students taking at least one course is $A \cup G$. By Theorem 5.1.1,

$$|A \cup G| = |A| + |G| - |A \cap G| = 30 + 25 - 8 = 47.$$

The 8 students enrolled in both courses were counted in $|A|$ and in $|G|$; subtracting 8 leaves each student

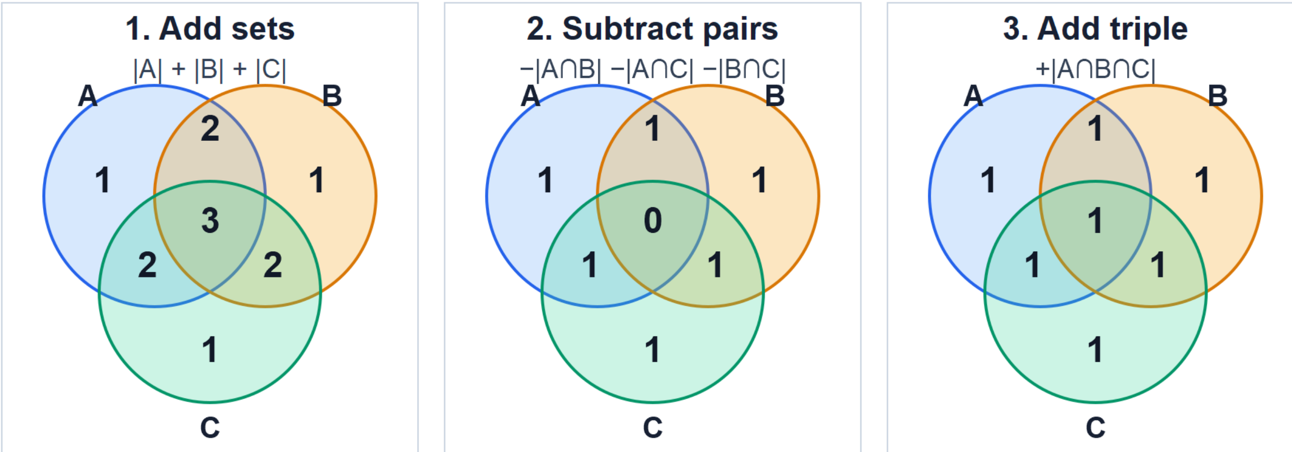

Figure 1: Multiplicities in each Venn region across the three stages of inclusion–exclusion for three sets. Stage 1 adds individual sizes; Stage 2 subtracts pairwise intersections; Stage 3 adds the triple intersection. *Figure description: Three side-by-side Venn diagrams for sets A, B, and C. In the first, regions belonging to exactly one, two, or three sets are labeled with multiplicities 1, 2, and 3. In the second, after subtracting pairwise intersections, they are labeled 1, 1, and 0. In the third, after adding the triple intersection, all seven regions inside the union are labeled 1.*

counted exactly once.

Now consider three finite sets A, B, and C. If we begin with the sum of their individual sizes, $|A|+|B|+|C|$, an element belonging to exactly one set is counted once. An element belonging to the intersection of two sets (say $A \cap B$, but not C) is counted twice: once in $|A|$ and once in $|B|$. An element belonging to all three sets, $A \cap B \cap C$, is counted three times.

To correct for the pairwise overlaps, we subtract the sizes of all three two-way intersections:

$$-|A \cap B| - |A \cap C| - |B \cap C|.$$

An element in exactly two sets was counted twice and is now subtracted once, leaving a net count of $2-1=1$. However, an element in all three sets was counted three times in the initial sum and subtracted three times (once in each pairwise intersection). Its net count is now $3-3=0$. To count such elements in the union, we must add back the size of the triple intersection $|A \cap B \cap C|$.

Figure 1 tracks how each step adjusts the multiplicity of every region in the Venn diagram. The net result is that every element belonging to at least one of the three sets has a final coefficient of 1.

Theorem 5.1.3 (Inclusion–exclusion for three sets). For any finite sets A, B, and C,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Example 5.1.4 (Course enrollments in three subjects). A survey of 114 mathematics majors records course enrollments as follows: 30 students take algebra, 25 take geometry, and 50 take calculus. Furthermore, 8 take algebra and geometry, 15 take algebra and calculus, 12 take geometry and calculus, and 5 take all three courses.

1. How many students take at least one of these three courses?

2. How many students take none of these three courses?

Solution: Let A, G, and C denote the sets of students taking algebra, geometry, and calculus, respectively.

1. The set of students taking at least one course is $A \cup G \cup C$. Applying Theorem 5.1.3:

$$\begin{aligned}|A \cup G \cup C| &= (|A| + |G| + |C|) - (|A \cap G| + |A \cap C| + |G \cap C|) + |A \cap G \cap C| \\ &= (30 + 25 + 50) - (8 + 15 + 12) + 5 \\ &= 105 - 35 + 5 = 75.\end{aligned}$$

2. The universal set U has $|U| = 114$ students. The students taking none of the three courses form the complement $U \setminus (A \cup G \cup C)$. By the difference rule,

$$|U \setminus (A \cup G \cup C)| = |U| - |A \cup G \cup C| = 114 - 75 = 39.$$

5.2 The general principle of inclusion–exclusion

To state the principle for any finite number of sets $A_1, A_2, \dots, A_m$, we index the intersections by subsets of the index set $\{1, 2, \dots, m\}$. The size $|I|$ determines whether the term is added or subtracted: terms with odd $|I|$ carry a positive sign, while terms with even $|I|$ carry a negative sign.

Theorem 5.2.1 (Principle of inclusion–exclusion)**.** Let $A_1, A_2, \dots, A_m$ be finite sets. Then

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_{\emptyset \neq I \subseteq \{1,2,\dots,m\}} (-1)^{|I|+1} \left|\bigcap_{i \in I} A_i\right|.$$

Grouping the index subsets by their size $k = |I|$ gives an equivalent expanded form:

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_{k=1}^{m} (-1)^{k+1} \sum_{\substack{I \subseteq \{1,\dots,m\} \\ |I| = k}} \left|\bigcap_{i \in I} A_i\right|.$$

The inner sum for $k = 1$ adds the sizes of all individual sets; for $k = 2$, it subtracts all $C(m, 2)$ pairwise intersections; for $k = 3$, it adds all $C(m, 3)$ triple intersections, continuing up to $k = m$.

5.2.1 Proof via indicator functions

A clean and general proof of Theorem 5.2.1 uses indicator functions. Let U be a finite universal set containing $A_1, \dots, A_m$. For any subset $A \subseteq U$, the **indicator function** $\mathbf{1}_A : U \to \{0, 1\}$ is defined by

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Indicator functions satisfy three basic algebraic properties:

1. The cardinality of A is the sum of its indicator values over the universe:
$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

2. The indicator of an intersection is the product of the indicators, since an element belongs to an intersection precisely when it belongs to every set in the intersection:
$$\mathbf{1}_{\bigcap_{i \in I} A_i}(x) = \prod_{i \in I} \mathbf{1}_{A_i}(x).$$

3. The indicator of the complement $U \setminus A$ is
$$\mathbf{1}_{U \setminus A}(x) = 1 - \mathbf{1}_A(x).$$

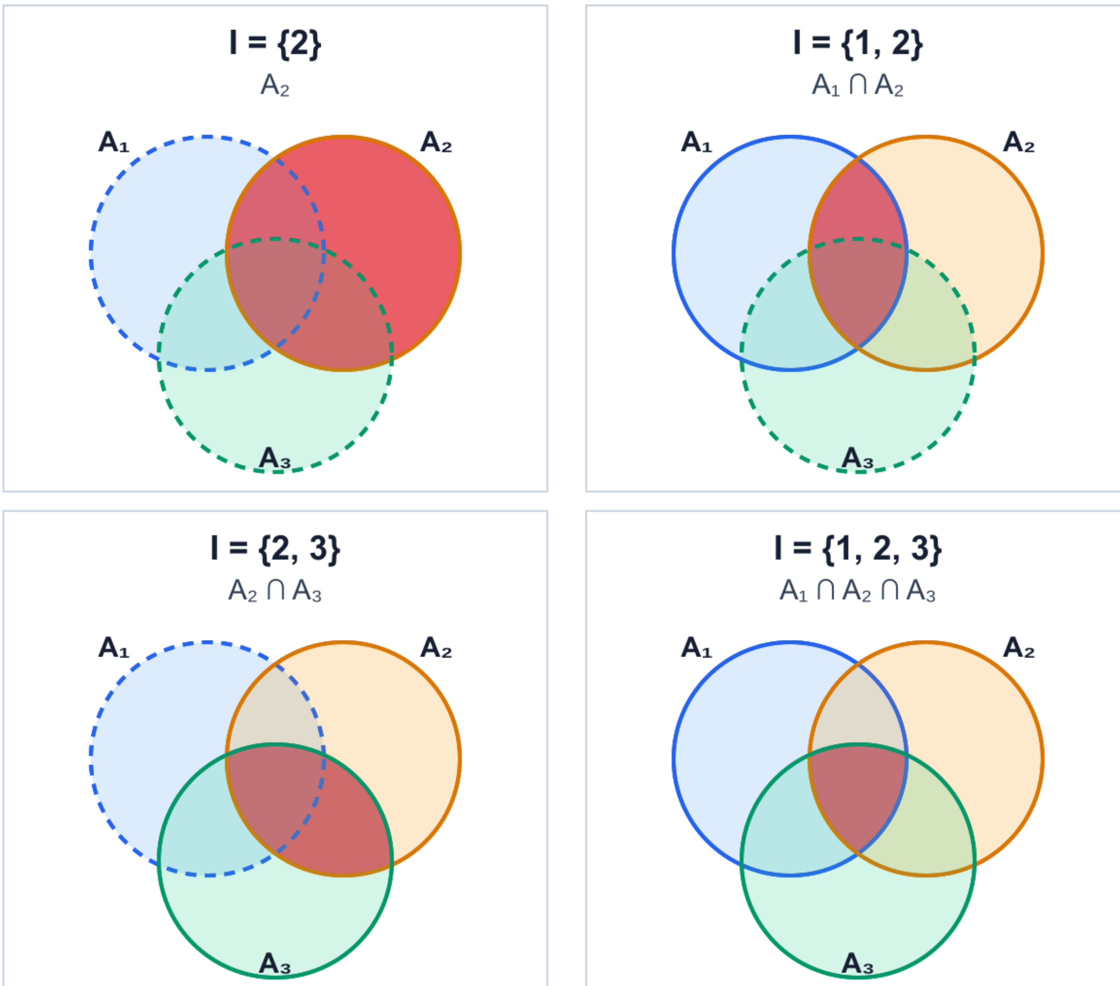

Figure 2: Four intersections corresponding to index subsets of $\{1, 2, 3\}$: $I = \{2\}$ gives A_2; $I = \{1, 2\}$ gives $A_1 \cap A_2$; $I = \{2, 3\}$ gives $A_2 \cap A_3$; and $I = \{1, 2, 3\}$ gives $A_1 \cap A_2 \cap A_3$.

Figure description: A two-by-two grid of Venn diagrams for three sets A1, A2, and A3, showing highlighted regions for A2, A1 intersect A2, A2 intersect A3, and the triple intersection A1 intersect A2 intersect A3.

Proof of Theorem 5.2.1. An element $x \in U$ does not belong to the union $\bigcup_{i=1}^{m} A_i$ if and only if $x \notin A_i$ for every $i \in \{1, \dots, m\}$. Therefore,

$$\mathbf{1}_{U \setminus \bigcup_{i=1}^{m} A_i}(x) = \prod_{i=1}^{m} (\mathbf{1}_{U \setminus A_i}(x)). = \prod_{i=1}^{m} (1 - \mathbf{1}_{A_i}(x)).$$

Using the complement property $\mathbf{1}_{\bigcup A_i}(x) = 1 - \mathbf{1}_{U \setminus \bigcup A_i}(x)$, we have

$$\mathbf{1}_{\bigcup_{i=1}^{m} A_i}(x) = 1 - \prod_{i=1}^{m} (1 - \mathbf{1}_{A_i}(x)).$$

We expand the product $\prod_{i=1}^{m} (1 - y_i)$ with $y_i = \mathbf{1}_{A_i}(x)$. Every term in the expansion corresponds to choosing $-y_i$ for a subset of indices $I \subseteq \{1, \dots, m\}$ and 1 for the remaining indices:

$$\prod_{i=1}^{m} (1 - \mathbf{1}_{A_i}(x)) = \sum_{I \subseteq \{1, \dots, m\}} (-1)^{|I|} \prod_{i \in I} \mathbf{1}_{A_i}(x) = 1 + \sum_{\emptyset \neq I \subseteq \{1, \dots, m\}} (-1)^{|I|} \mathbf{1}_{\bigcap_{i \in I} A_i}(x).$$

Subtracting this product from 1 yields

$$\mathbf{1}_{\bigcup_{i=1}^{m} A_i}(x) = \sum_{\emptyset \neq I \subseteq \{1, \dots, m\}} (-1)^{|I|+1} \mathbf{1}_{\bigcap_{i \in I} A_i}(x).$$

Finally, summing this identity over all $x \in U$ and using the identity $|B| = \sum_{x \in U} \mathbf{1}_B(x)$ for each term gives

$$\left| \bigcup_{i=1}^{m} A_i \right| = \sum_{\emptyset \neq I \subseteq \{1, \dots, m\}} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|,$$

which proves the theorem. $\square$

5.2.2 The complement and violation-set form

In combinatorial applications, we frequently want to count objects in a universe U that satisfy several conditions simultaneously. Rather than working directly with the satisfying objects, it is often simpler to identify the **violation sets** $A_1, A_2, \dots, A_m$, where $A_i \subseteq U$ consists of the objects that fail condition i.
An object satisfies all m conditions if and only if it belongs to none of the violation sets. The set of valid objects is therefore the complement of the union of the violation sets:

$$U \setminus \bigcup_{i=1}^{m} A_i.$$

Subtracting the union formula from $|U|$ gives the complement form of inclusion–exclusion:

$$\left| U \setminus \bigcup_{i=1}^{m} A_i \right| = |U| - \left| \bigcup_{i=1}^{m} A_i \right| = \sum_{I \subseteq \{1, \dots, m\}} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|, \tag{1}$$

where the term for $I = \emptyset$ is defined to be $|U|$.

5.2.3 Symmetric and grouped intersections

When all k-fold intersections have the same size, the summation simplifies substantially. If every intersection of k distinct sets has size N_k, then because there are $C(m, k)$ ways to choose an index subset of size k, the sum of all terms involving k sets is

$$S_k = \sum_{|I|=k} \left| \bigcap_{i \in I} A_i \right| = C(m, k) N_k.$$

In this case, the union and complement formulas become

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_{k=1}^{m}(-1)^{k+1}C(m,k)N_k,$$

$$\left|U \setminus \bigcup_{i=1}^{m} A_i\right| = |U| + \sum_{k=1}^{m}(-1)^{k}C(m,k)N_k = \sum_{k=0}^{m}(-1)^{k}C(m,k)N_k,$$

with $N_0 = |U|$.

Example 5.2.2 (Decimal strings containing a zero). How many decimal strings of length 4 contain at least one digit equal to 0?

Solution: Let U be the set of all $10^4 = 10000$ decimal strings of length 4. For $i \in \{1,2,3,4\}$, let A_i be the set of strings whose ith digit is 0. The set of strings containing at least one zero is $A_1 \cup A_2 \cup A_3 \cup A_4$. If we fix k specified positions to be 0, the remaining $4-k$ positions may each be filled by any of the 10 digits. Thus, every k-fold intersection has size

$$N_k = 10^{4-k},$$

regardless of which k positions are selected. By the grouped inclusion–exclusion formula,

$$\begin{aligned}|A_1 \cup A_2 \cup A_3 \cup A_4| &= C(4,1)10^3 - C(4,2)10^2 + C(4,3)10^1 - C(4,4)10^0 \\ &= 4(1000) - 6(100) + 4(10) - 1(1) \\ &= 4000 - 600 + 40 - 1 = 3439.\end{aligned}$$

We can verify this result by complementary counting: there are $9^4 = 6561$ strings that contain no zero digit, so $|U \setminus (A_1 \cup \cdots \cup A_4)| = 10000 - 6561 = 3439$.

5.3 Integer solutions with upper bounds

The method of stars and bars counts nonnegative integer solutions to equations of the form $x_1+x_2+\cdots+x_r = N$. Lower bounds $x_i \geq \ell_i$ are easily handled by a change of variables $x_i' = x_i - \ell_i$. Upper bounds $x_i \leq q_i$, however, cannot be incorporated by a simple variable shift because solutions may violate the bound in one coordinate, two coordinates, or more. Inclusion–exclusion provides the mechanism to enforce upper bounds.

Example 5.3.1 (Integer compositions with an upper bound). How many integer solutions satisfy

$$x_1 + x_2 + x_3 + x_4 = 30, \qquad 0 \leq x_i \leq 9 \quad \text{for each } i \in \{1,2,3,4\}?$$

Solution: Let U be the universe of all nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 = 30$, with no upper bounds imposed. By stars and bars,

$$|U| = C(30+4-1, 4-1) = C(33,3) = \frac{33 \cdot 32 \cdot 31}{6} = 5456.$$

For $i \in \{1,2,3,4\}$, let A_i be the set of solutions in U that violate the ith upper bound, which means $x_i \geq 10$. The solutions satisfying all upper bounds form the complement $U \setminus (A_1 \cup A_2 \cup A_3 \cup A_4)$.
To compute the size of a k-fold intersection $\bigcap_{i\in I} A_i$ for $|I| = k$, we require $x_i \geq 10$ for each $i \in I$. We make the substitution $x_i' = x_i - 10$ for $i \in I$ and $x_i' = x_i$ for $i \notin I$. The transformed variables x_i' are nonnegative integers satisfying

$$\sum_{i=1}^{4} x_i' = 30 - 10k.$$

By stars and bars, the number of such solutions is $C(30-10k+3, 3) = C(33-10k, 3)$ when $30-10k \geq 0$, and 0 when $30 - 10k < 0$.

- For $k = 1$: $N_1 = C(33 - 10, 3) = C(23, 3) = 1771$.
- For $k = 2$: $N_2 = C(33 - 20, 3) = C(13, 3) = 286$.
- For $k = 3$: $N_3 = C(33 - 30, 3) = C(3, 3) = 1$.
- For $k = 4$: $30 - 40 < 0$, so $N_4 = 0$.

Applying the complement formula:

$$\begin{aligned} |U \setminus (A_1 \cup A_2 \cup A_3 \cup A_4)| &= |U| - C(4,1)N_1 + C(4,2)N_2 - C(4,3)N_3 + C(4,4)N_4 \\ &= 5456 - 4(1771) + 6(286) - 4(1) + 0 \\ &= 5456 - 7084 + 1716 - 4 = 84. \end{aligned}$$

There is an instructive shortcut that confirms this result: let $y_i = 9 - x_i$. The conditions $0 \le x_i \le 9$ become $0 \le y_i \le 9$, and their sum is

$$y_1 + y_2 + y_3 + y_4 = 4(9) - (x_1 + x_2 + x_3 + x_4) = 36 - 30 = 6.$$

Since any nonnegative integers summing to 6 automatically satisfy $y_i \le 6 < 9$, the upper bounds are redundant for the y-variables. Stars and bars directly gives $C(6 + 4 - 1, 4 - 1) = C(9, 3) = 84$.

The same argument yields a general formula when all variables share a common upper bound.

Theorem 5.3.2 (Nonnegative solutions with a common upper bound)**.** Let r, N, q be nonnegative integers with $r \ge 1$. The number of integer solutions to

$$x_1 + x_2 + \cdots + x_r = N, \qquad 0 \le x_i \le q \quad \text{for } 1 \le i \le r,$$

is given by

$$\sum_{k=0}^{\min\{r, \lfloor N/(q+1) \rfloor\}} (-1)^k C(r, k) C(N - k(q + 1) + r - 1, r - 1).$$

Proof. Let U be the set of all nonnegative integer solutions to $\sum_{i=1}^{r} x_i = N$. For each $i \in \{1, \dots, r\}$, let $A_i = \{(x_1, \dots, x_r) \in U : x_i \ge q+1\}$. For any index subset $I \subseteq \{1, \dots, r\}$ of size k, setting $x_i' = x_i - (q+1)$ for $i \in I$ and $x_i' = x_i$ for $i \notin I$ gives a bijection between $\bigcap_{i \in I} A_i$ and the nonnegative solutions to $\sum_{i=1}^{r} x_i' = N - k(q + 1)$.
When $N - k(q+1) \ge 0$, stars and bars gives $C(N - k(q+1) + r - 1, r - 1)$ solutions. When $N - k(q+1) < 0$, the intersection is empty. Substituting these intersection counts into the complement form of inclusion–exclusion yields the claimed sum. □

5.4 Counting surjective functions and full-alphabet strings

Let X and Y be finite sets with $|X| = n$ and $|Y| = m$. A function $f : X \to Y$ is **surjective** (or **onto**) if every element of Y is in the image of f: for each $y \in Y$, there is at least one $x \in X$ such that $f(x) = y$.
Equivalently, viewing f as a string of length n whose characters are chosen from an alphabet Y of size m, surjectivity means that the string contains at least one copy of every symbol in Y.
Counting surjective functions directly is difficult because the number of ways to assign inputs depends on how the outputs are partitioned. However, the condition of *omitting* a specific element $y \in Y$ is easy to count: if y is excluded from the image, every $x \in X$ must map to one of the remaining $m - 1$ elements.

Theorem 5.4.1 (Number of surjective functions)**.** Let X and Y be finite sets with $|X| = n$ and $|Y| = m$, where $n, m \geq 1$. The number of surjective functions $f : X \to Y$ is

$$\sum_{k=0}^{m} (-1)^k C(m,k)(m-k)^n.$$

Proof. Let U be the set of all m^n functions from X to Y. For each $y \in Y$, let A_y be the set of functions whose image does not contain y:

$$A_y = \{f \in U : y \notin f(X)\}.$$

A function is surjective if and only if it belongs to none of the sets A_y.
For any subset $K \subseteq Y$ of size k, the intersection $\bigcap_{y \in K} A_y$ consists of functions whose image is contained in $Y \setminus K$. Since $|Y \setminus K| = m - k$, there are $(m-k)^n$ such functions.
There are $C(m,k)$ ways to choose the subset K of k omitted elements. By the complement form of inclusion–exclusion, the number of surjective functions is

$$\left| U \setminus \bigcup_{y \in Y} A_y \right| = \sum_{k=0}^{m} (-1)^k C(m,k)(m-k)^n,$$

which proves the formula. □

Example 5.4.2 (Surjective functions from a 5-set to a 3-set)**.** How many surjective functions are there from a set of 5 elements to a set of 3 elements?
Solution: Setting $n = 5$ and $m = 3$ in Theorem 5.4.1:

$$\begin{aligned}\sum_{k=0}^{3} (-1)^k C(3,k)(3-k)^5 &= C(3,0)3^5 - C(3,1)2^5 + C(3,2)1^5 - C(3,3)0^5 \\ &= 1 \cdot 243 - 3 \cdot 32 + 3 \cdot 1 - 1 \cdot 0 \\ &= 243 - 96 + 3 = 150.\end{aligned}$$

Remark 5.4.3 (The case $n < m$ and Stirling numbers)**.** When $n < m$, the pigeonhole principle implies that no function $f : X \to Y$ can be surjective, because n elements cannot cover $m > n$ targets. The formula correctly evaluates to 0 in this case due to exact algebraic cancellation among the terms of the alternating sum.
Furthermore, every surjective function $f : X \to Y$ partitions the domain X into m nonempty preimages $f^{-1}(y)$, and there are $m!$ ways to assign these m parts to the elements of Y. Denoting the number of partitions of an n-element set into m nonempty subsets by $S(n,m)$ (the **Stirling numbers of the second kind**), we have

$$S(n,m) = \frac{1}{m!} \sum_{k=0}^{m} (-1)^k C(m,k)(m-k)^n.$$

5.5 Permutations with forbidden substrings

When counting arrangements of distinct items that must avoid specific consecutive patterns, we combine the block-reduction technique with inclusion–exclusion.
To enforce that a specific substring appears, we treat that entire sequence of symbols as a single compound object. If the forbidden substrings do not share characters, computing the size of each intersection corresponds to gluing multiple blocks together.

Example 5.5.1 (Arrangements avoiding three disjoint blocks). How many permutations of the nine letters in `MODERNIST` contain none of the substrings `DO`, `RE`, or `MI`?

Solution: The word `MODERNIST` contains 9 distinct letters: `D, E, I, M, N, O, R, S, T`. The universe U consists of all $9! = 362880$ permutations.

We define three violation sets:

- A_1: permutations containing the substring `DO`.
- A_2: permutations containing the substring `RE`.
- A_3: permutations containing the substring `MI`.

We want to determine $|U \setminus (A_1 \cup A_2 \cup A_3)|$.

1. **Single violations ($k = 1$):** In A_1, we treat `DO` as a single block. Together with the remaining 7 individual letters, there are 8 objects to arrange, giving $|A_1| = 8!$. By symmetry, $|A_2| = |A_3| = 8!$.
2. **Pairwise intersections ($k = 2$):** The three substrings share no letters. For $A_1 \cap A_2$, both `DO` and `RE` are glued into blocks. Together with the remaining 5 letters, there are 7 objects, so $|A_1 \cap A_2| = 7!$. Similarly, $|A_1 \cap A_3| = |A_2 \cap A_3| = 7!$.
3. **Triple intersection ($k = 3$):** For $A_1 \cap A_2 \cap A_3$, all three blocks `DO`, `RE`, and `MI` are formed. Together with the remaining 3 letters (`N, S, T`), there are 6 objects to arrange, so $|A_1 \cap A_2 \cap A_3| = 6!$.

Applying inclusion–exclusion:

$$\begin{aligned}|U \setminus (A_1 \cup A_2 \cup A_3)| &= 9! - C(3,1)8! + C(3,2)7! - C(3,3)6! \\ &= 362880 - 3(40320) + 3(5040) - 720 \\ &= 362880 - 120960 + 15120 - 720 = 256320.\end{aligned}$$

Warning 5.5.2 (Overlapping and chained substrings). When forbidden substrings share letters (such as avoiding `AB` and `BC`), an intersection requires forming larger merged blocks (`ABC`). If two forbidden substrings are mutually contradictory (such as `AB` and `BA` in the same two positions), their intersection is empty. In such problems, one must carefully determine the valid block structures for each specific index set before applying the formula.

5.6 Derangements and fixed points

Let $\pi : \{1, 2, \dots, n\} \to \{1, 2, \dots, n\}$ be a permutation of the first n positive integers. An index $i \in \{1, 2, \dots, n\}$ is a **fixed point** of π if $\pi(i) = i$.

Definition 5.6.1 (Derangement). A **derangement** of $\{1, 2, \dots, n\}$ is a permutation with no fixed points; that is, $\pi(i) \neq i$ for all $i \in \{1, 2, \dots, n\}$. The number of derangements of an n-element set is denoted by D_n or $!n$.

A classic illustration is the hat-check problem: if n people check their hats and the hats are returned completely at random, a derangement is an assignment in which no person receives their own hat.

Theorem 5.6.2 (Formula for derangements). For any integer $n \geq 1$, the number of derangements of

$\{1, 2, \ldots, n\}$ is

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right).$$

By convention, $D_0 = 1$.

Proof. Let U be the set of all $n!$ permutations of $\{1, 2, \ldots, n\}$. For each $i \in \{1, \ldots, n\}$, let A_i be the set of permutations that fix element i:

$$A_i = \{\pi \in U : \pi(i) = i\}.$$

A permutation is a derangement if and only if it belongs to none of the sets A_i, so $D_n = |U \setminus \bigcup_{i=1}^{n} A_i|$. Let $I \subseteq \{1, \ldots, n\}$ with $|I| = k$. A permutation in the intersection $\bigcap_{i \in I} A_i$ fixes all k elements in I. The remaining $n - k$ elements may be permuted arbitrarily in $(n - k)!$ ways. Thus,

$$\left|\bigcap_{i \in I} A_i\right| = (n - k)!,$$

regardless of which k elements are chosen.

Since there are $C(n, k)$ subsets of size k, the complement form of inclusion–exclusion gives

$$D_n = \sum_{k=0}^{n} (-1)^k C(n, k)(n - k)!.$$

Using the identity $C(n, k)(n - k)! = \frac{n!}{k!(n-k)!}(n - k)! = \frac{n!}{k!}$, we factor out $n!$:

$$D_n = \sum_{k=0}^{n} (-1)^k \frac{n!}{k!} = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!},$$

which proves the formula. □

The table below lists the first several values of D_n alongside $n!$:

n	0	1	2	3	4	5	6
$n!$	1	1	2	6	24	120	720
D_n	1	0	1	2	9	44	265

Example 5.6.3 (Returning items at random). Six students place their phones into a basket, and the phones are redistributed uniformly at random so that each student receives one phone.

1. In how many ways can the phones be returned so that no student receives their own phone?
2. In how many ways can the phones be returned so that exactly two students receive their own phones?

Solution:

1. This is the number of derangements of 6 items:

$$\begin{aligned} D_6 &= 6! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right) \\ &= 720 \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720}\right) \\ &= 360 - 120 + 30 - 6 + 1 = 265. \end{aligned}$$

2. To have exactly two students receive their own phones, we first choose which two students receive their own phones in $C(6, 2)$ ways. The remaining 4 phones must then be deranged among the other 4 students so that no further student receives their own phone:

$$C(6, 2)D_4 = 15 \cdot 9 = 135.$$

The reasoning in part (2) of Example 5.6.3 immediately generalizes to count permutations with any specified number of fixed points.

Corollary 5.6.4 (Permutations with exactly k fixed points)**.** The number of permutations of $\{1, 2, \dots, n\}$ having exactly k fixed points (where $0 \le k \le n$) is

$$C(n, k)D_{n-k} = \frac{n!}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

Proof. There are $C(n, k)$ ways to choose the k elements that remain fixed. For each such choice, the remaining $n - k$ elements must be deranged among themselves to ensure that no other element is fixed. There are D_{n-k} such derangements. By the product rule, the total number of permutations is $C(n, k)D_{n-k}$. □

Remark 5.6.5 (Asymptotic probability of a derangement)**.** The probability that a uniformly chosen permutation of n elements is a derangement is

$$\frac{D_n}{n!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

As $n \to \infty$, this sum converges rapidly to the Taylor series for e^{-1}:

$$\lim_{n\to\infty} \frac{D_n}{n!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = e^{-1} \approx 0.367879.$$

Because the series alternates with terms decreasing monotonically to zero, $|D_n/n! - e^{-1}| \le \frac{1}{(n+1)!}$. Multiplying by $n!$ shows that $|D_n - n!/e| \le \frac{1}{n+1} \le \frac{1}{2}$ for all $n \ge 1$. Consequently, D_n is the nearest integer to $n!/e$ for every $n \ge 1$:

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

5.7 Summary of the inclusion–exclusion framework

To apply the principle of inclusion–exclusion systematically to an enumeration problem:

1. **Define the universe U:** Identify the ambient set of all possible configurations without the restrictive conditions, and verify that $|U|$ can be computed.

2. **Define the violation sets A_i:** Formulate each restrictive condition so that failing condition i corresponds to membership in $A_i \subseteq U$.

3. **Characterize the intersections:** For an arbitrary index subset $I \subseteq \{1, \dots, m\}$, determine what mathematical structure an object in $\bigcap_{i \in I} A_i$ possesses.

4. **Count the intersections:** Compute $|\bigcap_{i\in I} A_i|$. When symmetry ensures that all k-fold intersections have the same size N_k, group the $C(m,k)$ terms.

5. **Evaluate the alternating sum:** Substitute the intersection counts into the union formula or the complement formula, paying attention to the alternating signs.

5.8 Further reading

The course resources page links to the reference texts for this course.

- *Applied Combinatorics* by Tucker, Sections 7.1–7.4, provides additional examples of inclusion–exclusion, rook polynomials, and general derangement problems.
- *Discrete Mathematics: An Open Introduction* by Levin, Sections 3.3 and 3.8, covers Venn diagram bookkeeping, onto functions, and counting multisets with bounded multiplicities.
- *Counting Rocks!* by Schepers, Section 2.7, develops the algebraic indicator function approach.